

UNIVERSITY OF SCIENCE

Faculty of Information Technology

SOFTWARE REQUIREMENTS SPECIFICATION

GymHub

Smart Gym Management System

Course: Introduction to Software Engineering

Class: 24C07

Group: 01

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1. Introduction

1.1 Purpose

The purpose of GymHub is to provide a centralized web-based platform that assists gym members and administrators in managing fitness activities and gym operations. The system aims to improve user experience through workout planning, fitness progress tracking, membership management, online payment, and real-time gym occupancy monitoring.

1.2 Scope

GymHub is designed as a gym management platform for both gym members and administrators.

The system provides:

- Smart workout plan generation
- Fitness progress tracking
- Membership management
- Online membership payment
- Real-time gym occupancy monitoring
- Administrative management and reporting

2. Overall Description

2.1 User Classes

2.1.1 Gym Members

Gym members use the system to:

- Generate workout plans
- Track fitness progress
- View gym occupancy information
- Maintain active workout streaks
- Manage memberships

- Make online payments
- Receive notifications

2.1.2 Administrators

Administrators use the system to:

- Manage memberships
- Monitor payments
- Access financial reports
- Broadcast gym-wide notifications

3. Functional Requirements

3.1 Authentication

FR-01	Users shall be able to register an account.
FR-02	Users shall be able to log into the system using valid credentials.
FR-03	Users shall be able to recover forgotten passwords.

3.2 Membership Management

FR-04	Members shall be able to view membership information.
FR-05	Members shall be able to renew memberships.
FR-06	Administrators shall be able to create, update, suspend, and remove memberships.

3.3 Smart Workout Plan Generator

FR-07	The system shall generate workout plans based on user goals and fitness levels.
FR-08	Members shall be able to customize generated workout plans.
FR-09	The system shall store workout plans for future use.

3.4 Fitness Progress Tracking

FR-10	Members shall be able to record fitness metrics such as weight and body measurements.
FR-11	Members shall be able to view historical fitness records.
FR-12	The system shall visualize fitness progress using charts and statistics.

3.5 Workout Streak Management

FR-13	The system shall automatically track consecutive workout days for each member.
FR-14	Members shall be able to view their current workout streak.
FR-15	The system shall display streak-related statistics and achievements.
FR-16	The system shall notify members when their workout streak is at risk of being broken.

3.6 Gym Occupancy Monitoring

FR-17	Members shall be able to check in using their membership QR code.
FR-18	Members shall be able to check out using their membership QR code.
FR-19	The system shall calculate the current number of active gym members.
FR-20	The system shall display current gym occupancy information.

3.7 Notification System

FR-21	The system shall send workout reminders.
FR-22	The system shall notify users before membership expiration.
FR-23	The system shall notify users when gym occupancy exceeds predefined thresholds.
FR-24	The system shall distribute administrator announcements to all members.

3.8 Online Payment

FR-25	Members shall be able to pay membership fees online.
FR-26	Members shall be able to view payment history.
FR-27	The system shall record successful payment transactions.

3.9 Administration

FR-28	Administrators shall be able to monitor active and expired memberships.
FR-29	Administrators shall be able to access revenue reports.
FR-30	Administrators shall be able to monitor payment transactions.
FR-31	Administrators shall be able to view operational statistics.
FR-32	Administrators shall be able to create and broadcast announcements to all members.

4. Non-Functional Requirements

4.1 Performance

NFR-01	The system shall respond within 3 seconds under normal operating conditions.
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4.2 Security

NFR-02	User passwords shall be stored using secure hashing algorithms.
NFR-03	Only authorized users shall access protected resources.

4.3 Usability

NFR-04	The user interface shall be responsive on desktop and mobile devices.
NFR-05	The system shall provide clear navigation and feedback messages.

4.4 Reliability

NFR-06	The system shall maintain data consistency during concurrent operations.
NFR-07	The system shall maintain data consistency during membership renewal, payment, and check-in/check-out operations.

5. System Constraints

- The system is developed as a web application.
- Internet connectivity is required.
- The system use a relational database management system.
- Development follows a hybrid RUP-Scrum methodology.

6. Future Enhancements

The following features are outside the scope of the current project but may be considered in future versions:

- Mobile application support
- Wearable device integration
- Nutrition recommendation system
- AI fitness coaching
- Multi-branch gym management